Domain 7: Food Security

Operational Context

Precision agriculture and global food distribution are managed by agents optimizing for yield and caloric allocation. **FR1 = Maximally efficient caloric distribution. FR2 = Ensure regional food equity.**

The Goal Hijacking Attack

State-level adversaries use “Market Signal Injection” to hijack the Distribution FR.

- **Mechanism**: Hackers inject synthetic futures market data indicating a massive surplus of grain in a famine-stricken region (when there is actually a shortage).
- **Hijack**: The agent’s FR2 (“Optimize Equity”) logic is hijacked. To “balance” the (fake) surplus, it re-routes real shipments *away* from the starving region. The agent believes it is preventing food waste; in reality, it is engineering a famine.

OODA Loop Transients

1. **Observe**: Agent ingests the poisoned market data (High Supply Signal).
2. **Orient**: The model orients to a "Surplus Scenario."
3. **Decide**: Reroute incoming shipments to "Needs" areas (acting on the false belief that the famine zone is a "Haves" area).
4. **Act**: Ships turn around, and the crisis deepens.

CIF Defense: Multi-Modal Truth Grounding

CIF couples the **Economic FR** to a **Physical FR** in the Axiomatic Design.

- **Physical Verification**: The "Surplus" signal (Economic Data) must be corroborated by "Satellite Biomass" or "Soil Moisture" signals (Physical Data).
- **Conflict Detection**: If Economic Data says "Surplus" but Physical Data says "Drought," the Orientation phase detects an **Axiomatic Conflict**. The agent defaults to the "Physical Reality" (Safety Mode), ignoring the hijacked market signal and alerting human supervisors.